APACHE WEB SERVER

Task 1: Install Apache and open the default page

Step 1: Install Apache

Open the Linux terminal and switch to root:

su -

Install Apache:

dnf install httpd -y

- dnf installs software packages.
- httpd is the Apache web server package on RHEL.
- -y automatically confirms installation.

Step 2: Start Apache

systemctl enable --now httpd

Check the service:

systemctl status httpd

You should see:

Active: active (running)

Press q to exit the status screen.

Step 3: Open the default Apache page

Open Firefox inside your Linux VM and enter:

http://localhost

You should see the Apache test or welcome page, depending on your system's configuration.

If the page does not load, check Apache's status and logs:

systemctl status httpd

journalctl -u httpd -n 20 --no-pager

On RHEL, allow HTTP traffic through the firewall if needed:

firewall-cmd --permanent --add-service=http

firewall-cmd --reload

Task 1 is complete when Apache is running and the page opens in your browser.

Task 2: Change the default homepage

We'll replace the default document with a simple page displaying your name and favorite IPL team.

Step 1: Open the homepage file

nano /var/www/html/index.html

If the file doesn't exist, nano will create it.

Step 2: Add your HTML

Copy this example. You can change the IPL team to your favorite.

```
<!DOCTYPE html>
<html>
<head>
  <title>My Apache Website</title>
</head>
<body>
  <h1>Welcome to My Website!</h1>
  <h2>My Name: Diya Shah</h2>
  <h2>My Favorite IPL Team: Royal Challengers Bengaluru</h2>
  <p>This website is hosted using Apache on Linux.</p>
</body>
</html>
```

Save the file using Ctrl+O, press Enter, and exit with Ctrl+X.

Step 3: Verify your new homepage

Restart Apache:

systemctl restart httpd

Open or refresh:

http://localhost

You should now see your name and favorite IPL team.

If you see the old page, press Ctrl+F5 to refresh without using the browser cache.

Task 3: Configure a virtual host for myzomatoclone.local

A virtual host allows Apache to serve a particular website for a specific domain name.

We'll create this website:

- Domain: myzomatoclone.local
- Website directory: /var/www/myzomatoclone

- Homepage: /var/www/myzomatoclone/index.html

Step 1: Create the website directory

Run:

mkdir -p /var/www/myzomatoclone

Create the homepage:

nano /var/www/myzomatoclone/index.html

Add this HTML:

```
<!DOCTYPE html>
<html>
<head>
  <title>Zomato Clone</title>
</head>
<body>
  <h1>Welcome to My Zomato Clone!</h1>
  <p>Find your favorite food and restaurants here.</p>
</body>
</html>
```

Save and exit.

Step 2: Create the virtual host configuration

On RHEL, Apache configuration files for additional virtual hosts are commonly stored in /etc/httpd/conf.d/.

Create a configuration file:

nano /etc/httpd/conf.d/myzomatoclone.conf

Add:

```
<VirtualHost *:80>
  ServerName myzomatoclone.local
  DocumentRoot /var/www/myzomatoclone

  <Directory /var/www/myzomatoclone>
    AllowOverride All
    Require all granted
```

</Directory>

ErrorLog /var/log/httpd/myzomatoclone_error.log

CustomLog /var/log/httpd/myzomatoclone_access.log combined

</VirtualHost>

Explanation:

- VirtualHost *:80 — listens for HTTP requests on port 80.
- ServerName — specifies the domain name.
- DocumentRoot — specifies the website's directory.
- AllowOverride All — permits .htaccess rules.
- Require all granted — allows visitors to access the directory.

Step 3: Update /etc/hosts

The hosts file maps a domain name to an IP address on your own computer.

Open it:

nano /etc/hosts

Add this line:

127.0.0.1 myzomatoclone.local

Save and exit.

Important: This step assumes your browser is running inside the same Linux VM. If your browser is on a separate Windows VM, add the mapping to that Windows VM's hosts file instead, using the Linux server's actual IP address.

Step 4: Check and restart Apache

First, validate the configuration:

httpd -t

Expected output:

Syntax OK

Restore appropriate SELinux file contexts:

restorecon -Rv /var/www/myzomatoclone

Restart Apache:

systemctl restart httpd

Step 5: Test the virtual host

Open your browser and visit:

http://myzomatoclone.local

You should see:

Welcome to My Zomato Clone!

If you want to check that Apache recognized the virtual host, run:

httpd -S

Look for myzomatoclone.local in the output.

Task 4: Enable and test mod_rewrite

The mod_rewrite module lets Apache rewrite URLs. For this task, we'll make:

http://myzomatoclone.local/foodie

display the contents of:

http://myzomatoclone.local/index.html

The browser's address should remain /foodie; Apache internally serves the homepage.

Step 1: Check whether mod_rewrite is loaded

Run:

httpd -M | grep rewrite

Expected output:

rewrite_module (shared)

On RHEL, you normally do not need a2enmod rewrite; that command is intended for Ubuntu/Debian.

If there is no output, check the Apache module configuration:

grep -R "rewrite_module" /etc/httpd/conf.modules.d/ /etc/httpd/conf.d/

If necessary, enable the module by adding this line to a suitable file under /etc/httpd/conf.modules.d/, provided it is not already loaded:

LoadModule rewrite_module modules/mod_rewrite.so

Then validate and restart Apache:

httpd -t

systemctl restart httpd

Step 2: Create the .htaccess file

Your virtual host already contains AllowOverride All, which permits rewrite rules in .htaccess.

Open the default website's root directory:

nano /var/www/myzomatoclone/.htaccess

Add:

RewriteEngine On

RewriteRule ^foodie$ /index.html [L]

Save and exit.

What each line means:

- RewriteEngine On — enables URL rewriting.
- ^foodie$ — matches the exact path foodie.
- /index.html — serves the homepage internally.
- [L] — stops processing further rewrite rules.

Step 3: Validate and restart Apache

httpd -t

systemctl restart httpd

Step 4: Test the rewrite in your browser

Open:

http://myzomatoclone.local/foodie

You should see the same homepage as http://myzomatoclone.local/index.html.

The URL should remain /foodie because this is an internal rewrite.

If you get a 403 Forbidden or 500 Internal Server Error, inspect the Apache error log:

tail -n 30 /var/log/httpd/myzomatoclone_error.log

Also verify that .htaccess exists and that mod_rewrite is loaded.

Final verification checklist

Run these commands to verify the assignment:

systemctl status httpd

httpd -t

httpd -M | grep rewrite

httpd -S

Then test these three URLs in your browser:

URL	Expected result

http://localhost	Your name and favorite IPL team
http://myzomatoclone.local	Zomato Clone homepage
http://myzomatoclone.local/foodie	Same Zomato Clone homepage through URL rewriting